

Long Term Water Use Sustainability

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Problem Definition

Climate change is well-known to pose an imminent threat to the availability of water supplies in many areas of the world. In particular, parts of the Southwest United States are expected to face water shortages in the near future. The awareness of these challenges has taken on renewed life amidst the recent proliferation of data centres across the region. We are interested in investigating the sustainability of water use and future water availability in the area.

We selected Maricopa County in Arizona, which contains the city of Phoenix, as a well-defined area of interest. Maricopa is the fourth-largest county in the US by population, being home to approximately 4.4 million people. At the same time, only 0.3% of the land area in Maricopa is covered by water. Potential future water shortages in this region present a real danger to a heavily populated area.

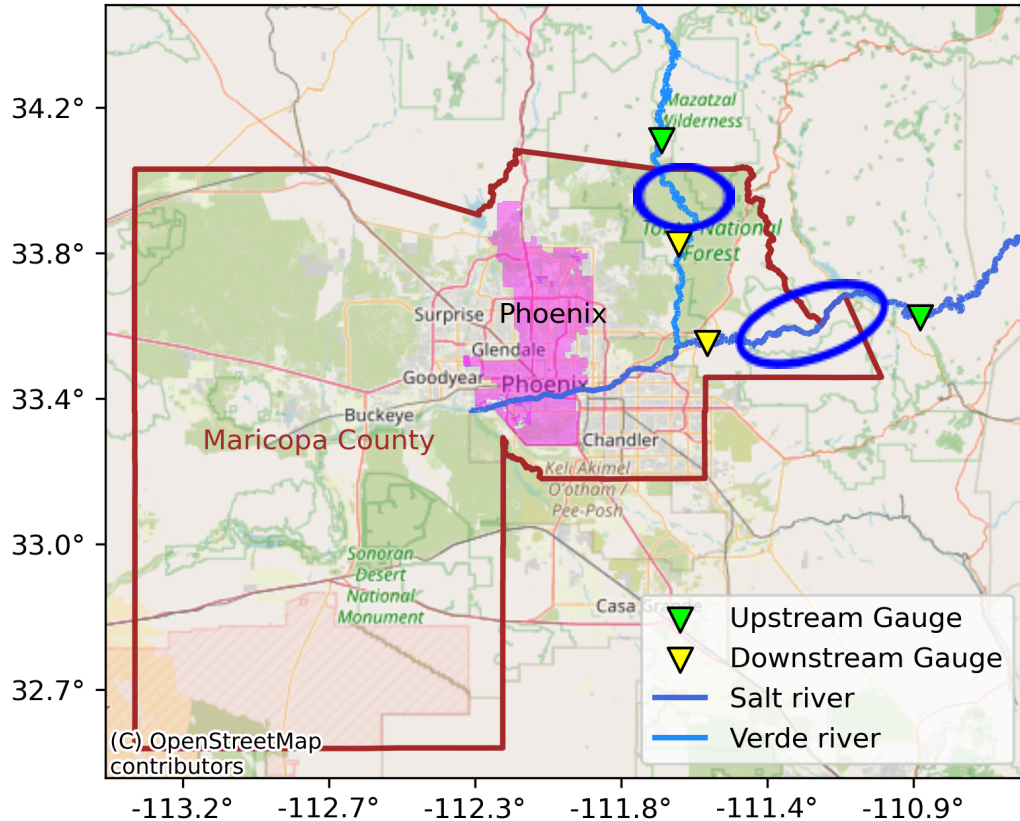
Phoenix receives most of its municipal water from the Salt River Project (SRP), which manages a system of reservoirs on the Salt River and Verde River. We aim to predict future changes in the water storage of the SRP using river inflow and outflow data, climate data, and human contributions to water use.

Data gathering

Response variables

Although we were interested in modelling reservoir water levels, data on that was not directly available. Instead, we obtained data on daily river flow rates (in cubic feet per second) from USGS water gauges. For each river, we chose one gauge just upstream of the reservoir, to estimate inflow into reservoirs, and one just downstream, to estimate outflow into reservoirs. Change in reservoir water storage can be estimated from these gauges by taking the cumulative difference between upstream and downstream flow.

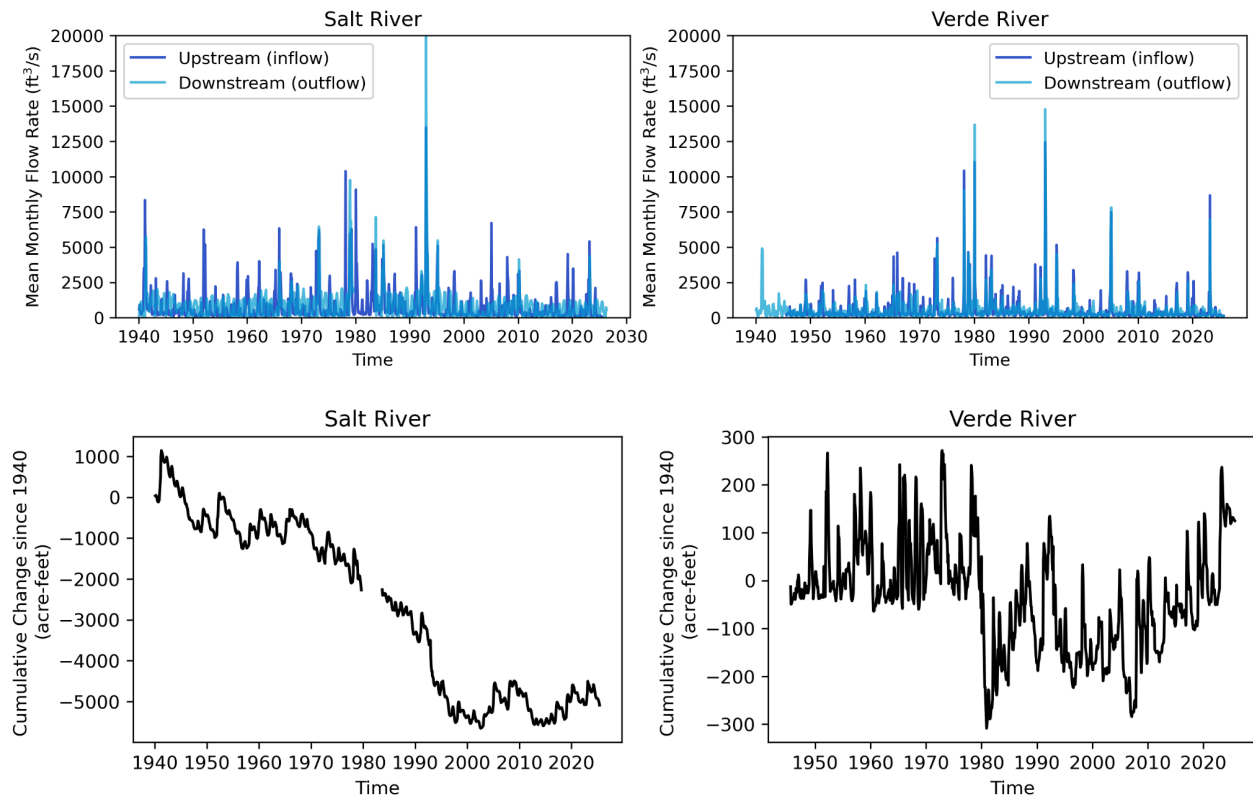
Daily flow data was aggregated to monthly timescales by taking the mean. Since the data had large outliers, it was log-transformed prior to analysis.



Map showing locations of river gauges used in this study. Blue circles show the approximate locations of reservoirs.

Details of response data

Variable name	USGS gage name	USGS gage number	Earliest data available
Salt river upstream flow rate	Salt River Near Roosevelt, AZ	USGS-09498500	1913-10-01
Salt river downstream flow rate	Salt River Below Stewart Mountain Dam, AZ	USGS-09502000	1934-10-01
Verde river upstream flow rate	Verde River Below Tangle Creek, Above Horseshoe Dam, AZ	USGS-09508500	1945-08-21
Verde river downstream flow rate	Verde River Below Bartlett Dam, AZ	USGS-09510000	1904-01-01



Temporal trends in river flow rates. Raw river flow data (top two panels) does not show any clear trends over time. However, by taking the cumulative difference between upstream and downstream flow (bottom two panels; estimate of change in reservoir water storage), water levels in the Salt river reservoirs (bottom left) appear to have declined over time.

Features

We were interested in modeling upstream water flow as a function of climate, and downstream flow as a function of human demand for water. Therefore, our features included climate (temperature and precipitation), population of Maricopa county, irrigated farmland area in Maricopa county, and number and capacity of datacenters in Maricopa county. The last three of these are considered to be proxies for human water demand. Climate data available was at monthly timescales, while the rest of the features were only available at annual or lower time resolution. They were therefore linearly interpolated to monthly timescales. Wherever possible, we also obtained data on future projections of features, in order to predict future values of river flow (future projections were found for all features except irrigated farmland area).

Details of feature data gathering

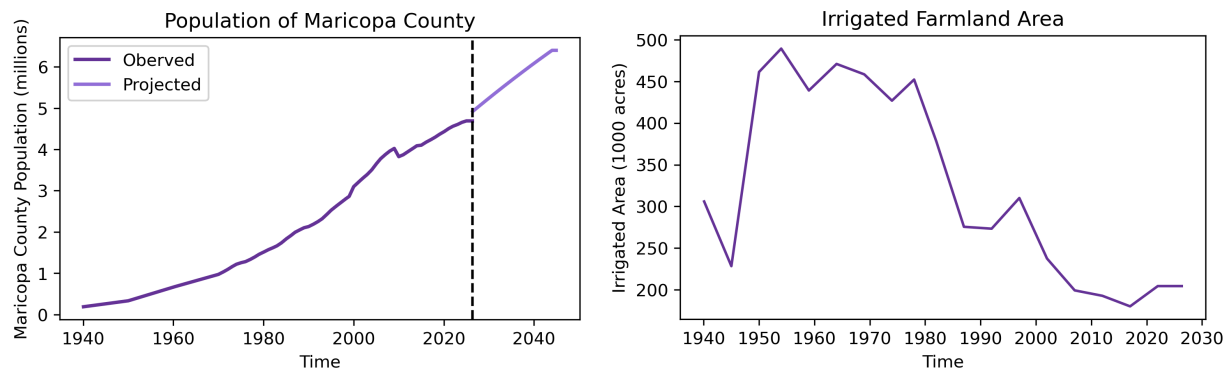
Variable name	Justification for inclusion	Data source
Mean monthly temperature in river watershed (separately for each river)	Higher the temperature, greater the water loss due to evaporation	<u>Historic data:</u> NOAA Monthly U.S. Climate Gridded Dataset (NClimGrid), monthly data from Jan 1, 1895 - May 1, 2026. Downloaded data for the entire US, then clipped to county boundaries or watershed boundaries, and then averaged temperature and summed precipitation for each watershed/county. <u>Future projections:</u> LOCA2 downscaled CMIP6 climate projections based on the ACCESS-CM2 climate model. Downloaded monthly data for entire North America, from 2015 - 2044, then summarized over areas of interest (watershed and county boundaries).
Mean monthly temperature in Maricopa county	Water demand may increase with temperature	
Total monthly precipitation in river watershed (separately for each river)	River water ultimately comes from precipitation	
Mean monthly precipitation in Maricopa county	Demand for river water may decrease with increasing precipitation (due to, for example, less water needed for lawns, crops etc., plus higher availability of local ground water)	
Population of Maricopa county	Proxy for municipal water demand	1920-1970 (decadal data): National Historical GIS 1970-2014 (annual data): National Bureau of Economic Research - U.S. Intercensal County Population Data, 1970-2014 2015-2025 (annual data): US Census Bureau - Intercensal Population Totals: 2010-2020; Annual Estimates of the Resident Population for Counties: April 1, 2020 to July 1, 2025 Future population projections - Arizona Office of Economic Statistics County Population Projections 2025 - 2060
Irrigated farmland area in Maricopa	Proxy for agricultural water demand	USDA Census of Agriculture, 1935-2022; conducted every 5 years. Accessed

county		census reports from the USDA National Agriculture Statistics Service
Number of data centers in Maricopa county	Data centers are known to use a lot of water	Cleanview.co (includes both currently operational and upcoming/planned datacenters)
Total MW capacity of data centers in Maricopa county		

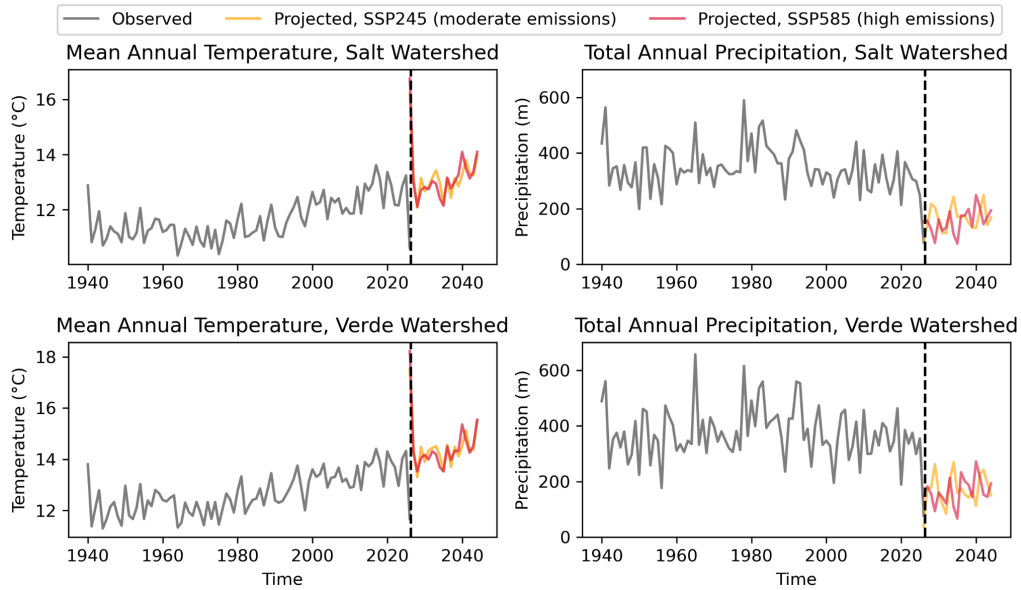
Other data needed for feature preprocessing

- Geographic boundaries of river watershed (used for climate data preprocessing) - USGS StreamStats Basin Delineation tool (<https://streamstats.usgs.gov/ss/>)
- Geographic boundaries of Arizona counties (used to get the boundaries of Maricopa county for climate data preprocessing) - Arizona Department of Health Services GIS Portal (<https://geodata-adhsgis.hub.arcgis.com/datasets/ADHSGIS::county-boundaries-in-arizona>)

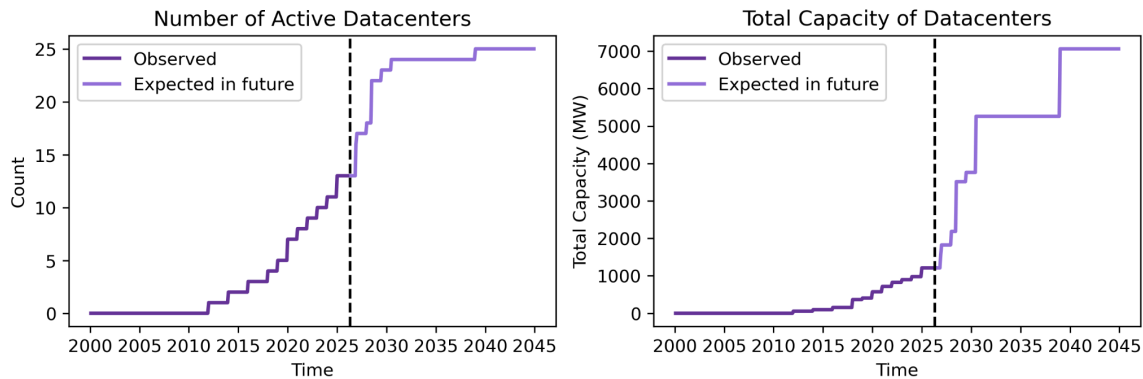
Temporal trends in features



Temporal trends in population and irrigated farmland area in Maricopa county. Population size has increased over time and is expected to continue increasing, which may increase municipal demand for water. Irrigated farmland area has declined in recent decades, which may be associated with a decline in agricultural water demand.



Climate in river watersheds over time. Average temperatures have increased and are expected to increase further in future. Precipitation is expected to decline in future. This could negatively affect water supply.



The number of datacenters in Maricopa county has increased rapidly in recent years, and several more are being built. This may increase the demand for water.

Model Selection

Summary

In this project, we model four target variables: Verde River upstream and downstream flow, and Salt River upstream and downstream flow. For each target variable, we selected features and performed hyperparameter tuning for several different models, and selected the features + hyperparameters + model combination that minimized RMSE in five-fold time series cross validation on our training data.

Verde River Upstream Flow

Models Tested:

- Random Forest
- AdaBoost
- Gradient Boost
- XGBoost.

Starting List of Features:

- Verde River watershed temperature
- Verde River watershed precipitation
- Verde River upstream flow lags 1-12 months.

Step 0: Data Cleaning and Train/Test Split

Data Cleaning:

Because the river flow data is liable to have occasional large spikes, we applied $\log(x + 1)$ to all river flow data.

Train/Test Split:

Full data for our starting list of features runs from August 1946 to September 2025. For model selection, we used August 1946 to November 2017 as our training set and December 2017 to September 2025 as our test set.

Step 1: Feature Selection

We wanted to select the top two most relevant lags for each of our four models. To do this we ran five-fold time series cross validation and selected the two lags that minimized average RMSE for each model across the five folds. The results of this are summarized in the following table.

Model	Selected Lags	Average Fold RMSE
Random Forest	1 & 7	0.4247
AdaBoost	1 & 2	0.5769
Gradient Boost	1 & 11	0.4154
XGBoost	1 & 4	0.4679

Step 2: Hyperparameter Tuning

We performed hyperparameter tuning for our four models, with the selected best lags for each. We did this with a custom grid search with five-fold time series cross validation. The hyperparameters tested are summarized below.

Model	Hyperparameters Tuned	Respective Ranges
Random Forest	max_depth	range(1,11)
	n_estimators	[100, 500]
AdaBoost	n_estimators	[50, 100, 200]
	learning_rate	[0.01, 0.1]
Gradient Boost	max_depth	range(1,11)
	n_estimators	[100, 500]
XGBoost	max_depth	range(1,11)
	n_estimators	[100, 500]

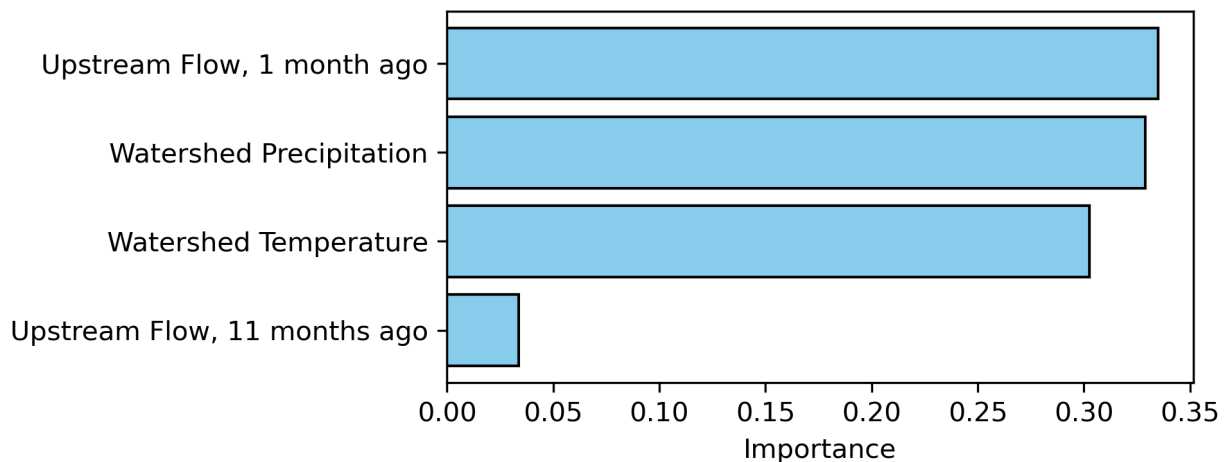
We selected the hyperparameters for each model that minimized average RMSE across the five folds. The results are summarized below.

Model	Selected Hyperparameters	Average Fold RMSE
Random Forest	- max_depth = 7 - n_estimators = 100	0.419
AdaBoost	- n_estimators = 50 - learning_rate = 0.2	0.4953
Gradient Boost	- max_depth = 3 - n_estimators = 100	0.4154
XGBoost	- max_depth = 1 - n_estimators = 100	0.4432

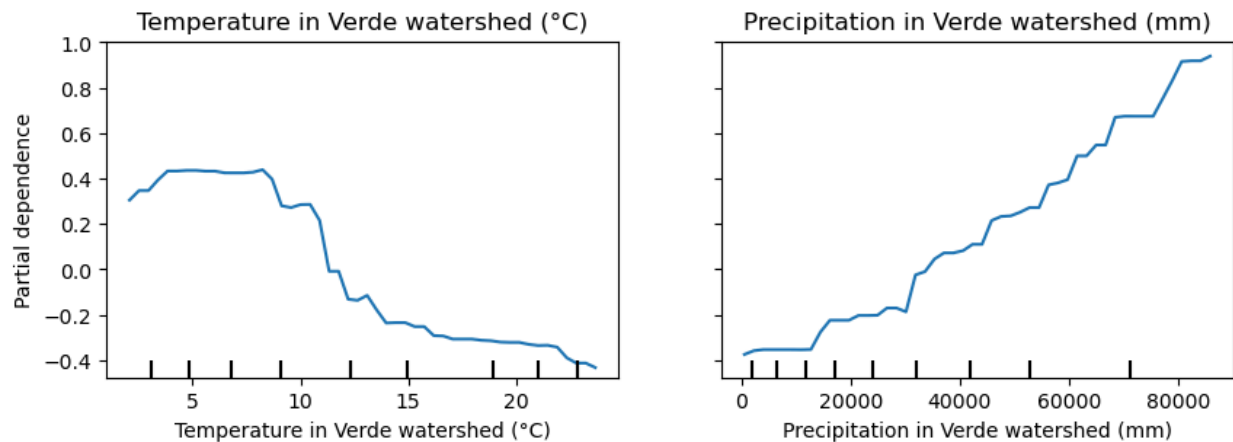
Step 3: Investigate Final Model

Our final selected model is Gradient Boost with lags 1 and 11, max_depth = 3 and n_estimators = 100. Feature importance for this model is as follows.

Feature	Importance
Verde upstream lag 1	0.334
Verde watershed precipitation	0.329
Verde watershed temperature	0.303
Verde upstream lag 11	0.033



Partial dependence plots are as follows.

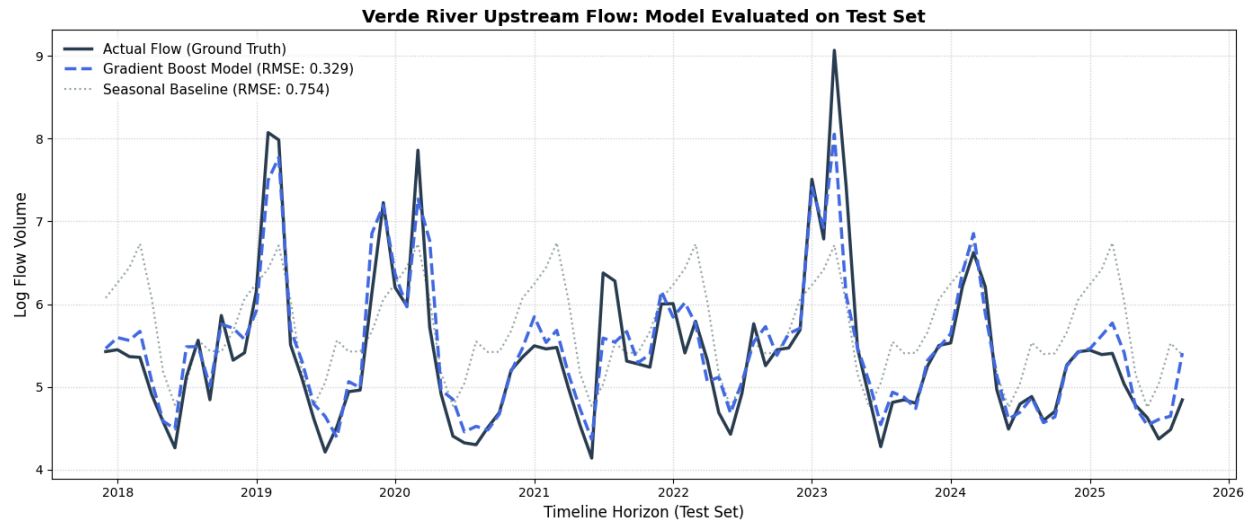


Step 4: Evaluate Final Model

To evaluate the performance of our final selected model, we generated predictions for our test set using one-month-ahead forecasting, and evaluated the RMSE of our generated predictions. We compared against a baseline Seasonal Average model. The results are summarized below.

Model	RMSE	% of Mean Log Flow
Gradient Boost (selected model)	0.3291	6.09%
Seasonal Average (baseline)	0.7539	13.95%

From the table we see that our selected Gradient Boost model reduces forecast RMSE by 56.35% relative to the Seasonal Average baseline. Plots of the selected Gradient Boost model predictions and Seasonal Average baseline predictions, together with the true log flow values, are below.



Verde River Downstream Flow

Models Tested:

- Random Forest
- ADA Boost
- Gradient Boost
- XGBoost.

Starting List of Features:

- Verde River watershed temperature
- Verde River watershed precipitation
- Maricopa County temperature
- Maricopa County precipitation
- Maricopa County population
- Maricopa County irrigated farmland
- Data centers total MW capacity
- Data centers total number
- Verde River upstream flow
- Verde River downstream flow lags 1-12 months.

Step 0: Data Cleaning and Train/Test Split

Data Cleaning:

Because the river flow data is liable to have occasional large spikes, we applied $\log(x + 1)$ to all river flow data.

Train/Test Split:

Full data for our starting list of features runs from August 1945 to September 2025. For model selection, we used August 1945 to November 2017 as our training set and December 2017 to September 2025 as our test set.

Step 1: Feature Selection

We wanted to select the top two most relevant lags for each of our four models. To do this we ran five-fold time series cross validation and selected the two lags that minimized average RMSE for each model across the five folds. The results of this are summarized in the table below.

Model	Selected Lags	Average Fold RMSE
Random Forest	4 & 12	0.9960
AdaBoost	1 & 6	1.0170
Gradient Boost	2 & 4	0.9756
XGBoost	5 & 6	0.9596

Step 2: Hyperparameter Tuning

We performed hyperparameter tuning for our four models, with the selected best lags for each. We did this with a custom grid search with five-fold time series cross validation. The hyperparameters tested are summarized below.

Model	Hyperparameters Tuned	Respective Ranges
Random Forest	max_depth	range(1,11)
	n_estimators	[100, 500]
AdaBoost	n_estimators	[50, 100, 200]
	learning_rate	[0.01, 0.1]
Gradient Boost	max_depth	range(1,11)
	n_estimators	[100, 500]
XGBoost	max_depth	range(1,11)
	n_estimators	[100, 500]

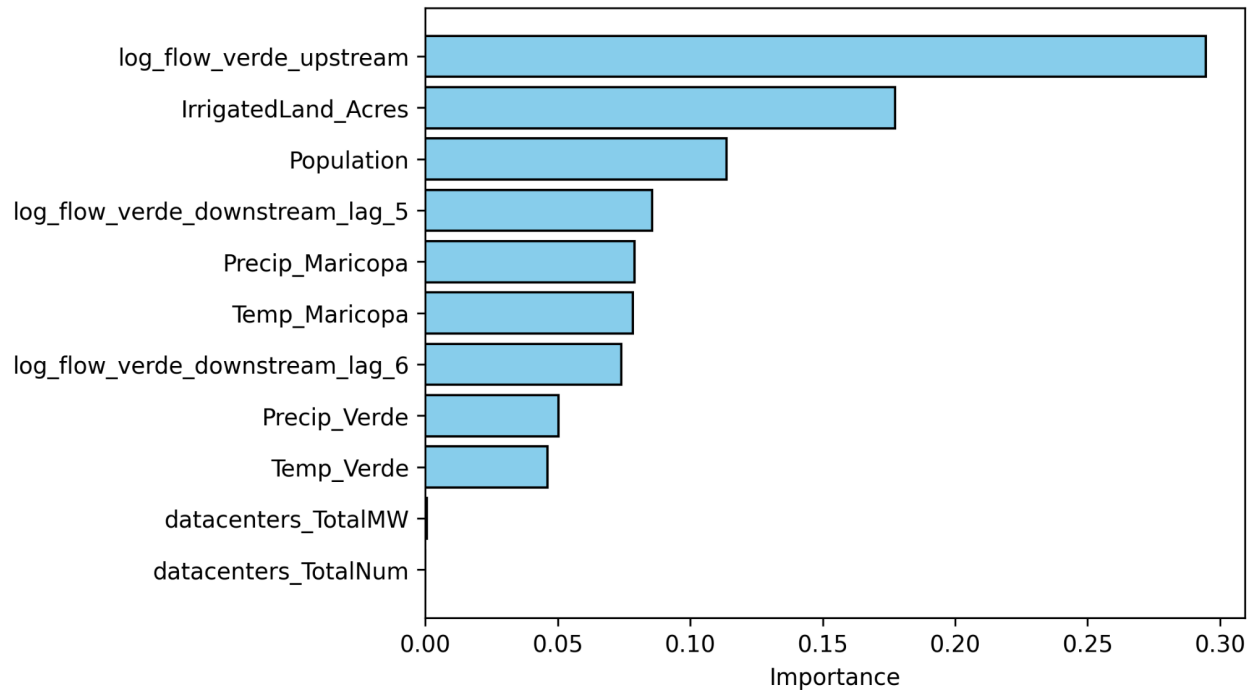
We selected the hyperparameters for each model that minimized average RMSE across the five folds. The results are summarized below.

Model	Selected Hyperparemeters	Average Fold RMSE
Random Forest	- max_depth = 10 - n_estimators = 500	0.9838
AdaBoost	- n_estimators = 200 - learning_rate = 0.1	0.9972
Gradient Boost	- max_depth = 3 - n_estimators = 100	0.9756
XGBoost	- max_depth = 6 - n_estimators = 100	0.9596

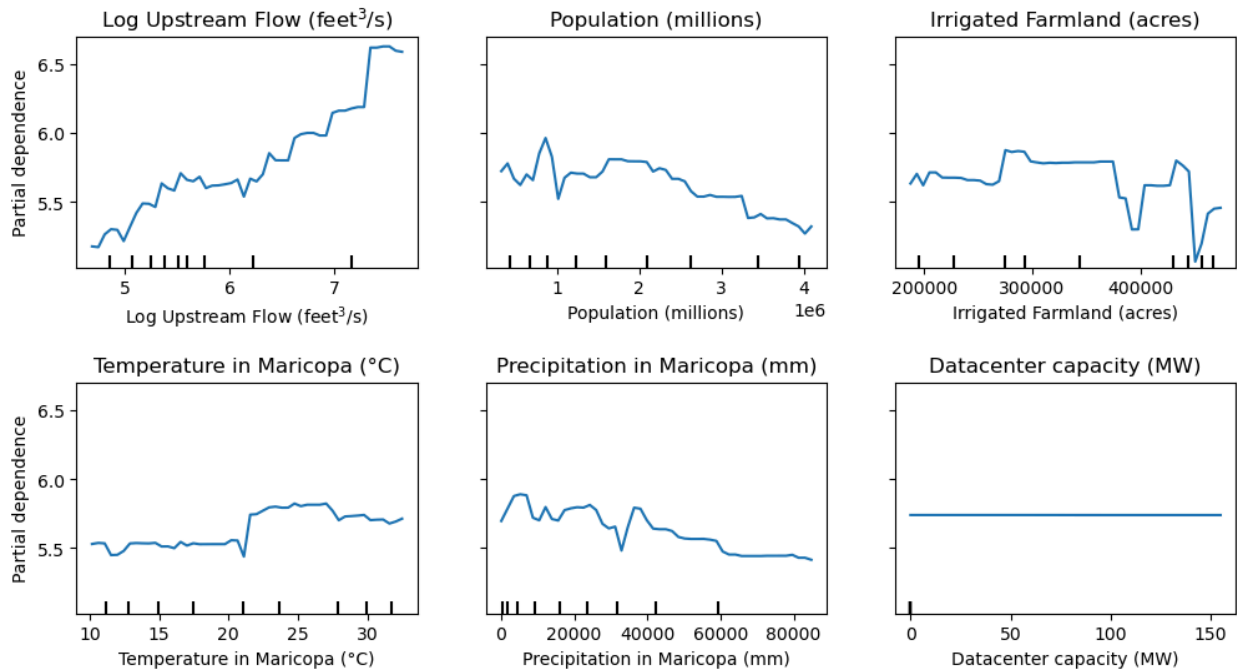
Step 3: Investigate Final Model

Our final selected model is XGBoost with lags 5 and 6, max_depth = 6 and n_estimators = 100. Feature importance for this model is as follows.

Feature	Importance
Verde River Upstream Flow	0.295
Maricopa County Irrigated Farmland	0.177
Maricopa County Population	0.114
Verde River Downstream Lag 5	0.086
Maricopa County Precipitation	0.079
Maricopa County Temperature	0.078
Verde River Downstream Lag 6	0.074
Verde River Watershed Precipitation	0.050
Verde River Watershed Temperature	0.046
Data Centers Total MW Capacity	0.001
Data Centers Total Number	0



Partial dependence plots are as follows.



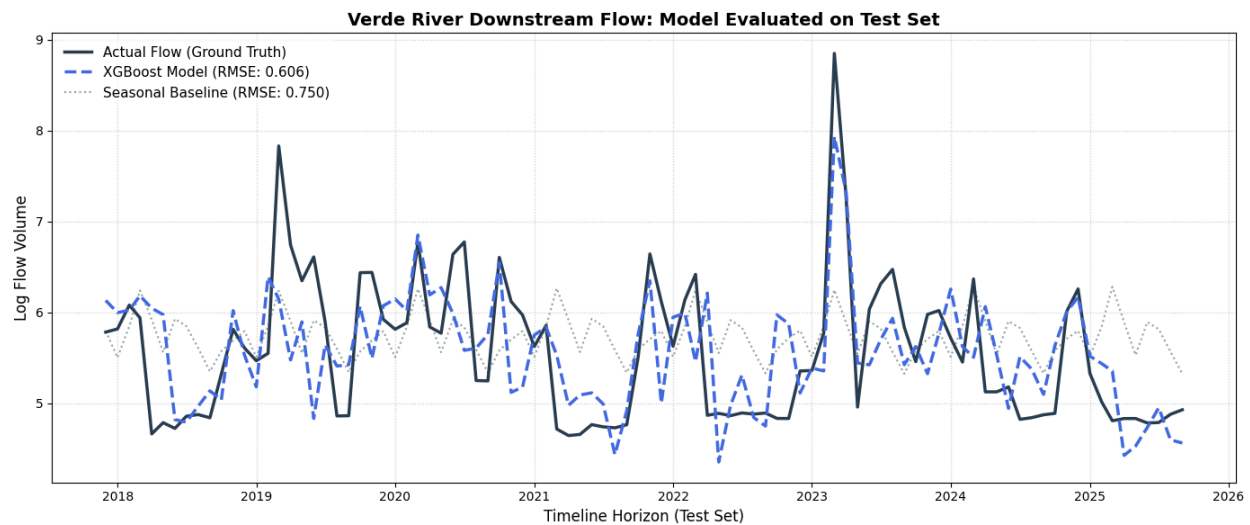
Interestingly, increasing population seems to decrease water demand, as seen in the partial dependency. This is likely because the residents of Phoenix have become more water efficient in recent years, resulting in decreasing water-use per person as population increased.

Step 4: Evaluate Final Model

To evaluate the performance of our final selected model, we generated predictions for our test set using one-month-ahead forecasting, and evaluated the RMSE of our generated predictions. We compared against a baseline Seasonal Average model. The results are summarized below.

Model	RMSE	% of Mean Log Flow
XGBoost (selected model)	0.6056	10.88%
Seasonal Average (baseline)	0.7500	13.47%

From the table we see that our selected XGBoost model reduces forecast RMSE by 19.25% relative to the Seasonal Average baseline. Plots of the selected Gradient Boost model predictions and Seasonal Average baseline predictions, together with the true log flow values, are below.



Salt River Upstream Flow

Models Tested:

- Random Forest
- XGBoost

Starting List of Features:

- Salt River watershed temperature
- Salt River watershed precipitation
- Salt River upstream flow lags 1-12 months

Step 0: Data Cleaning and Train/Test Split

Data Cleaning:

Because the river flow data is liable to have occasional large spikes, we applied $\log(x + 1)$ to all river flow data.

Train/Test Split:

Full data for our starting list of features runs from October 1914 to June 2025. For model selection, we used October 1914 to November 2017 as our training set and December 2017 to June 2025 as our test set.

Step 1: Feature Selection

We wanted to select the top two most relevant lags for each of our two models. To do this we ran three-fold time series cross validation and selected the two lags that minimized average RMSE for each model across the three folds. The results of this are summarized in the table below.

Model	Selected Lags	Average Fold RMSE
Random Forest	1 & 12	0.5959
XGBoost	1 & 2	0.6678

Step 2: Hyperparameter Tuning

We performed hyperparameter tuning for our two models, with the selected best lags for each. We did this with a custom grid search with three-fold time series cross validation. The hyperparameters tested are summarized below.

Model	Hyperparameters Tuned	Respective Ranges
Random Forest	max_depth	[4, 6, 8, 10]
	n_estimators	[100, 300]
XGBoost	max_depth	[4, 6, 8, 10]
	n_estimators	[100, 300]

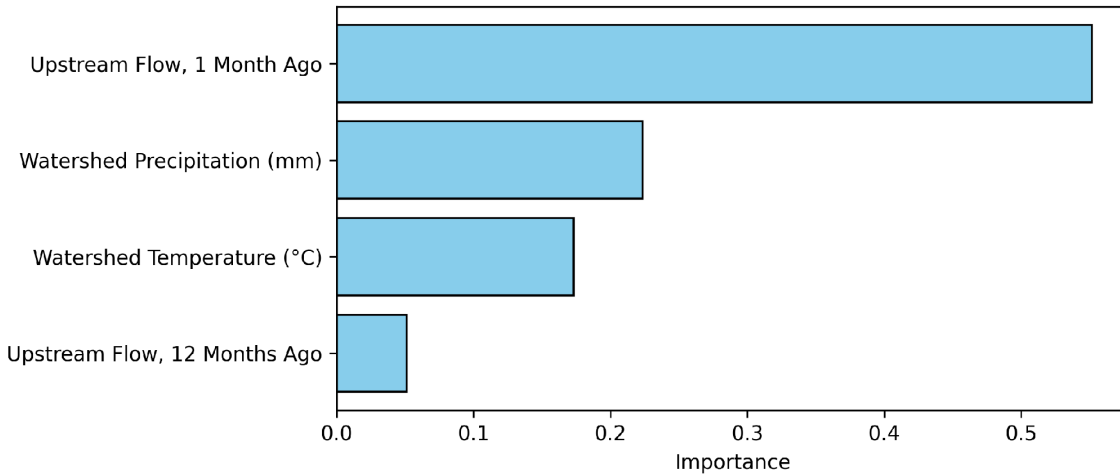
We selected the hyperparameters for each model that minimized average RMSE across the three folds. The results are summarized below.

Model	Selected Hyperparameters	Average Fold RMSE
Random Forest	max_depth = 10, n_estimators = 300	0.5693
XGBoost	max_depth = 4, n_estimators = 100	0.6621

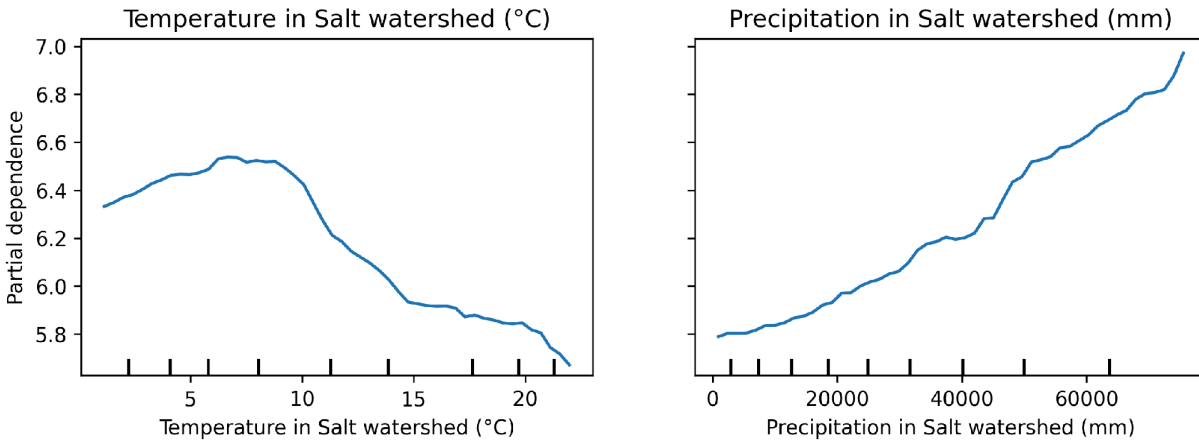
Step 3: Investigate Final Model

Our final selected model is Random Forest with lags 1 and 12, max_depth = 10 and n_estimators = 300. Feature importance for this model is as follows.

Feature	Importance
Salt upstream flow, lag 1	0.552
Salt watershed precipitation	0.224
Salt watershed temperature	0.173
Salt upstream flow, lag 12	0.051



Partial dependence plots are as follows.

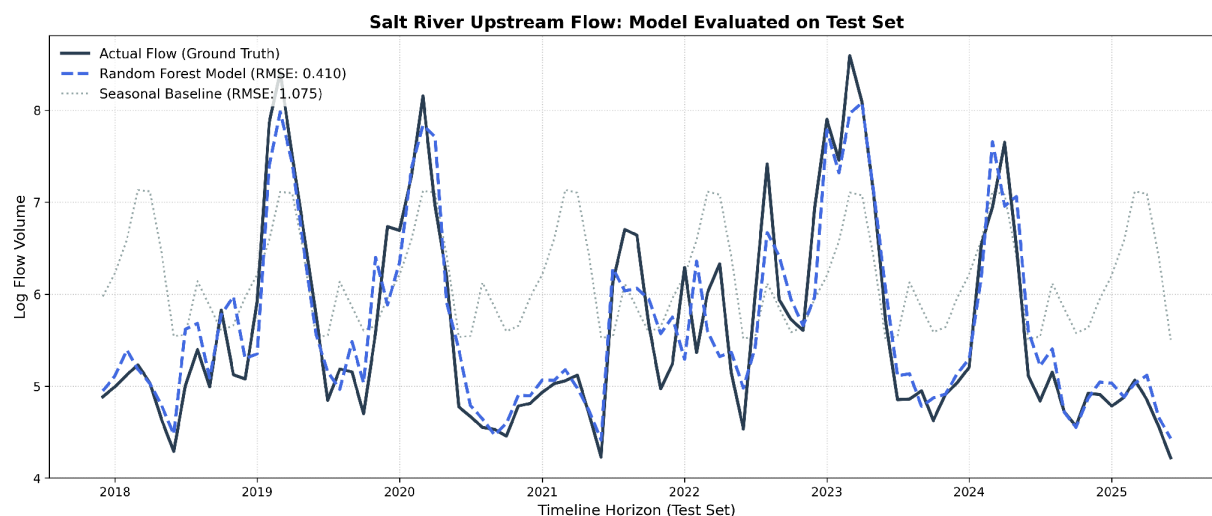


Step 4: Evaluate Final Model

To evaluate the performance of our final selected model, we generated predictions for our test set using one-month-ahead forecasting, and evaluated the RMSE of our generated predictions. We compared against a baseline Seasonal Average model. The results are summarized below.

Model	RMSE	% of Mean Log Flow
Random Forest (selected model)	0.4173	7.40%
Seasonal Average (baseline)	1.0752	19.07%

From the table we see that our selected Random Forest model reduces forecast RMSE by 61.19% relative to the Seasonal Average baseline. Plots of the selected Random Forest model predictions and Seasonal Average baseline predictions, together with the true log flow values, are below.



Salt River Downstream Flow

Models Tested:

- Random Forest
- XGBoost

Starting List of Features:

- Salt River watershed temperature
- Salt River watershed precipitation
- Maricopa County temperature
- Maricopa County precipitation
- Maricopa County population
- Maricopa County irrigated farmland
- Data centers total MW capacity
- Data centers total number
- Salt River upstream flow
- Salt River downstream flow lags 1-12 months

Step 0: Data Cleaning and Train/Test Split

Data Cleaning:

Because the river flow data is liable to have occasional large spikes, we applied $\log(x + 1)$ to all river flow data.

Train/Test Split:

Full data for our starting list of features runs from October 1935 to June 2025. For model selection, we used October 1935 to November 2017 as our training set and December 2017 to June 2025 as our test set.

Step 1: Feature Selection

We wanted to select the top two most relevant lags for each of our two models. To do this we ran three-fold time series cross validation and selected the two lags that minimized average RMSE for each model across the three folds. The results of this are summarized in the table below.

Model	Selected Lags	Average Fold RMSE
Random Forest	9 & 12	1.7355
XGBoost	12 & 2	1.7462

Step 2: Hyperparameter Tuning

We performed hyperparameter tuning for our two models, with the selected best lags for each. We did this with a custom grid search with three-fold time series cross validation. The hyperparameters tested are summarized below.

Model	Hyperparameters Tuned	Respective Ranges
Random Forest	max_depth	[4, 6, 8, 10]
	n_estimators	[100, 300]
XGBoost	max_depth	[4, 6, 8, 10]
	n_estimators	[100, 300]

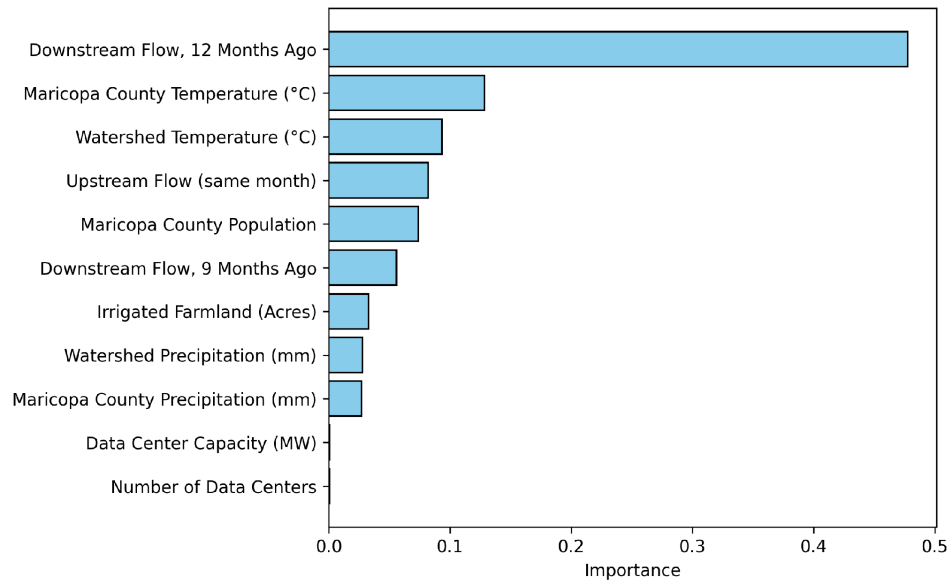
We selected the hyperparameters for each model that minimized average RMSE across the three folds. The results are summarized below.

Model	Selected Hyperparameters	Average Fold RMSE
Random Forest	max_depth = 6, n_estimators = 100	1.7355
XGBoost	max_depth = 6, n_estimators = 300	1.7441

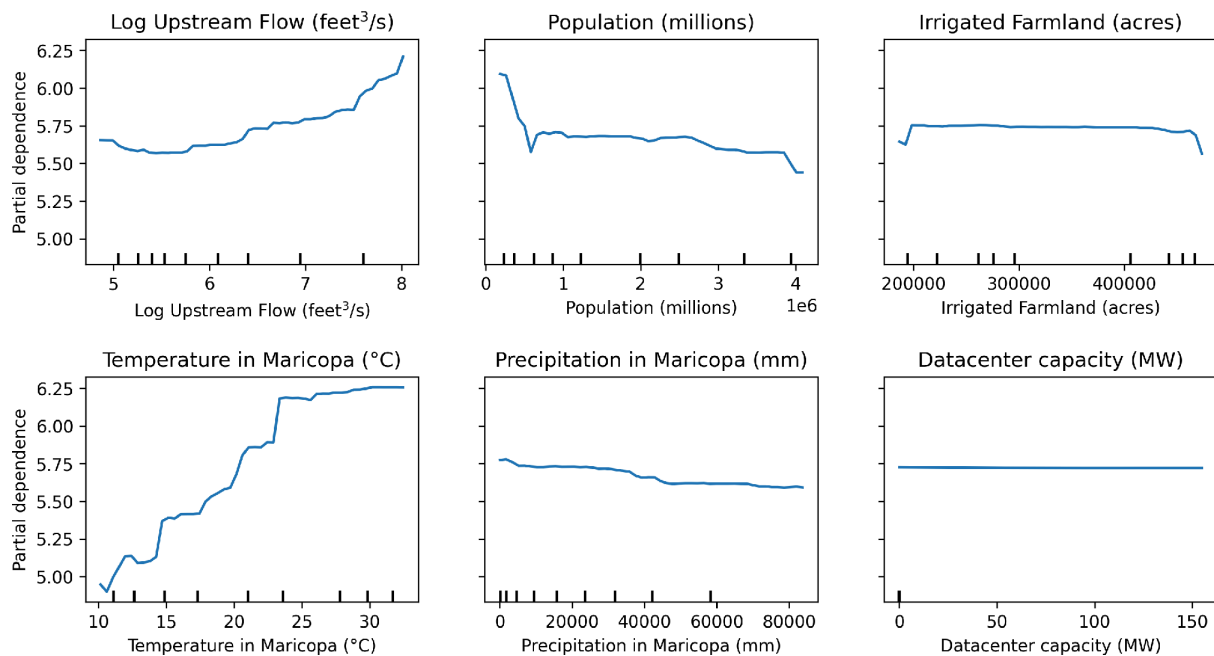
Step 3: Investigate Final Model

Our final selected model is Random Forest with lags 9 and 12, max_depth = 6 and n_estimators = 100. Feature importance for this model is as follows.

Feature	Importance
Salt downstream flow, lag 12	0.478
Maricopa County temperature	0.128
Salt upstream flow (current month)	0.082
Salt watershed temperature	0.093
Maricopa County population	0.074
Salt downstream flow, lag 9	0.056
Maricopa County irrigated farmland	0.033
Maricopa County precipitation	0.027
Salt watershed precipitation	0.028
Data centers total MW capacity	0.0007
Data centers total number	0.0005



Partial dependence plots are as follows.

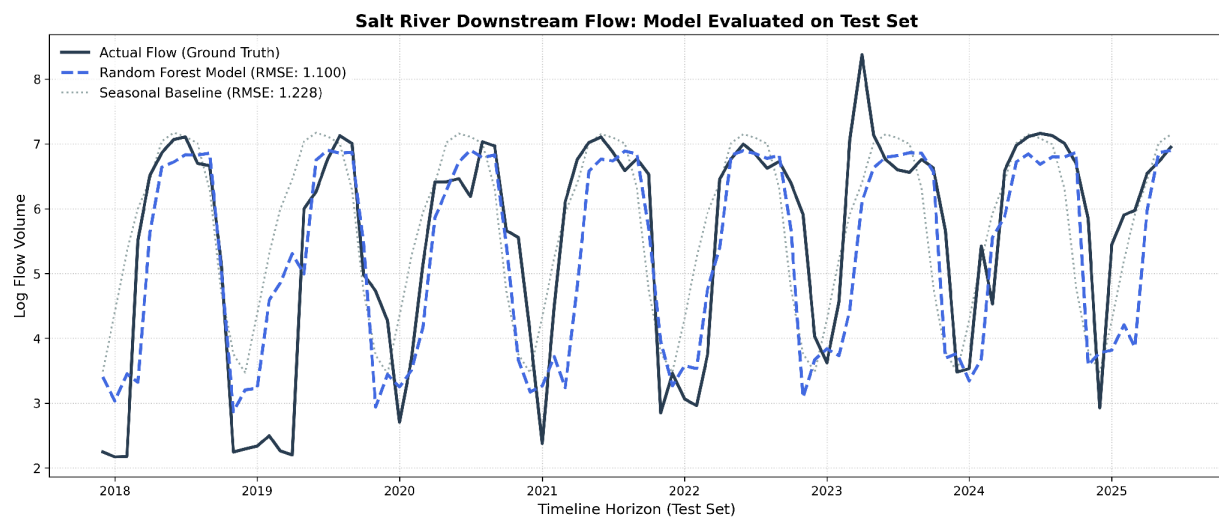


Step 4: Evaluate Final Model

To evaluate the performance of our final selected model, we generated predictions for our test set using one-month-ahead forecasting, and evaluated the RMSE of our generated predictions. We compared against a baseline Seasonal Average model. The results are summarized below.

Model	RMSE	% of Mean Log Flow
Random Forest (selected model)	1.1044	20.08%
Seasonal Average (baseline)	1.2282	22.33%

From the table we see that our selected Random Forest model reduces forecast RMSE by 10.08% relative to the Seasonal Average baseline. Plots of the selected Random Forest model predictions and Seasonal Average baseline predictions, together with the true log flow values, are below.

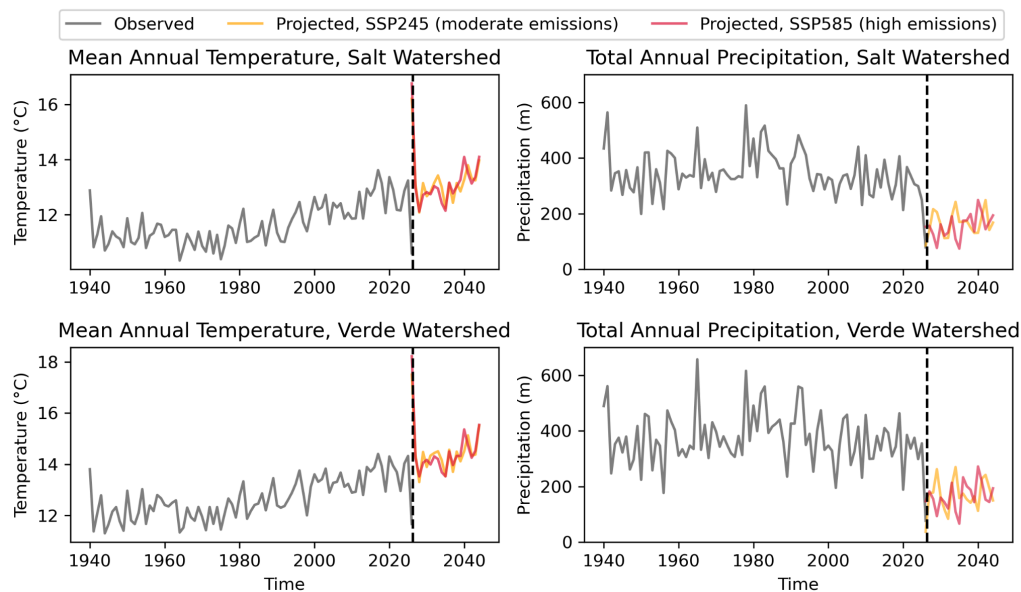


Future flow rate predictions

In order to predict river flow rates into the future, we require future climate data. We used the Shared Socio-economic Pathways (SSP) projections, which (among many other things) provide temperature and precipitation estimations in the Salt and Verde watersheds into the future.

The SSP presents projections for five scenarios based on a range of future climate action outcomes. For our analysis, we selected two:

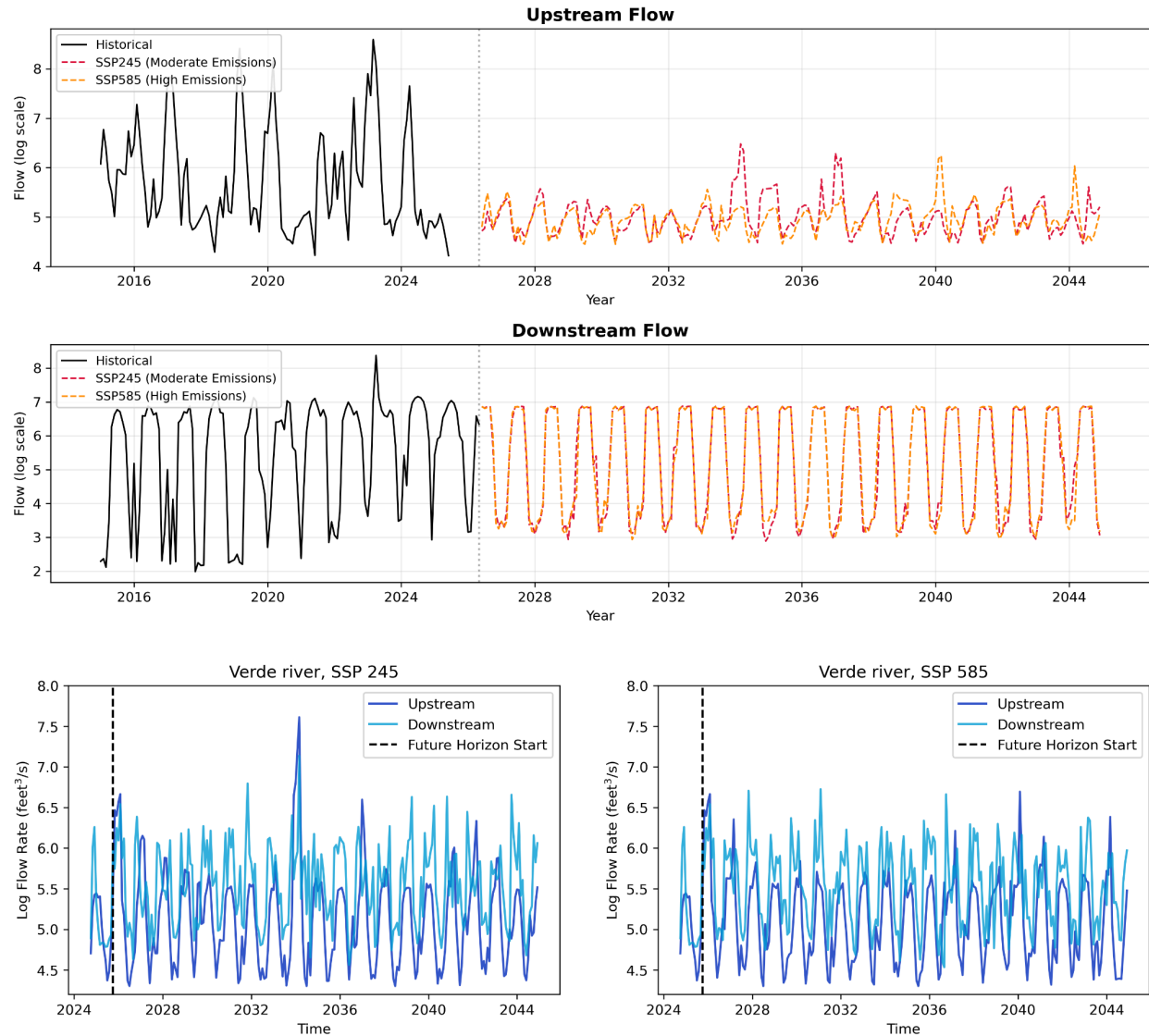
- SSP2-4.5, a moderate emissions scenario representing a “middle of the road” approach to climate change mitigation in the future
- SSP5-8.5, a high emissions scenario representing a future approach with continuing fossil fuel development



Projected annual temperature and precipitation for the Salt and Verde watersheds from the SSP2-4.5 and SSP5-8.5 scenarios

Using the selected models described in the previous sections, we predicted future monthly log flow rates under the SSP2-4.5 and SSP5-8.5 scenarios.

Salt River Flow: Historical and Projected Future



Predicted future log flow rates for the Salt and Verde rivers for two SSP climate projection scenarios. These are used to estimate changes to the SRP reservoir levels in the future. Notably the future outflow exceeds the future inflow for both rivers in both scenarios.

Uncertainty estimation for future reservoir change predictions

In order to estimate the uncertainties in the future predicted log monthly flow rates, we used a block bootstrapping method.

For each model, we generated 1000 block bootstrap resamples of the feature data using a block size of 25 months. Optimal block size is a current area of research, and the best available information suggested a block size roughly equal to the cube root of the total number of elements, and also one greater than a multiple of the period of the data. Since we know our period to be 12 months exactly, this led to a choice of 25 for the block size.

Each bootstrap resample was used to train the winning model and then used to predict the target flow data using the original feature data. In this way we obtained a sample of 1000 predictions for each month for the target flow in order to estimate the distribution of possible predictions, and estimate the uncertainty in the model's prediction ability.

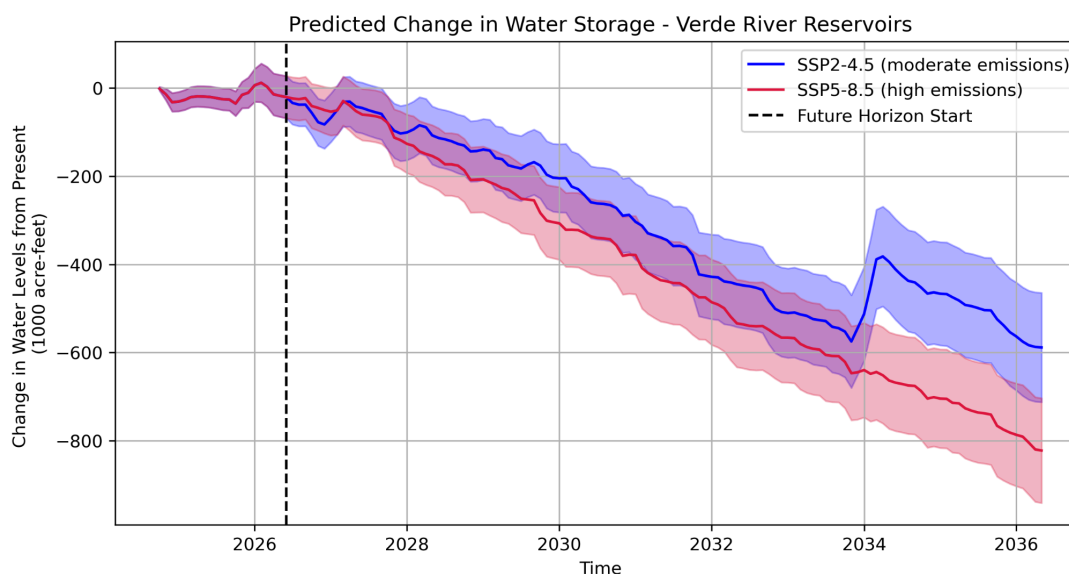
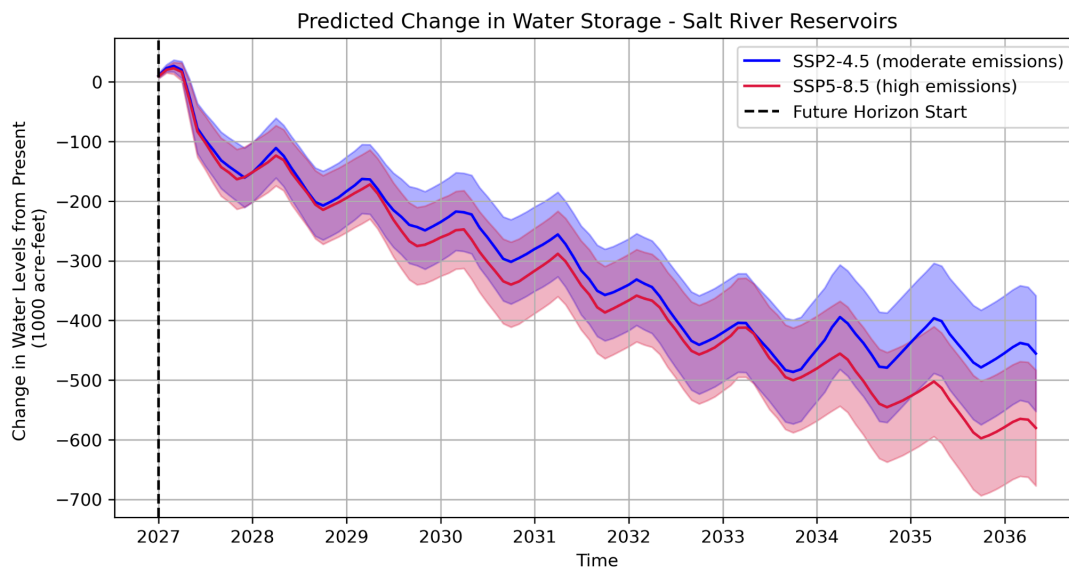
The calculation to obtain an uncertainty for the model as a whole was done by taking the 3rd quartile value among the individual prediction uncertainties, which is then taken to be the uniform uncertainty for each prediction. We opted to choose a generous estimate for the prediction for several reasons:

- Since the change in reservoir levels is the difference of the inflow and outflow, it is possible that there are systematics affecting the prediction that are not understood. This is an interesting area for further investigation
- As predictions progress into the future, the predictions are based on feature data that are themselves predicted rather than measured. Since it is not understood how uncertainties in the model inputs propagate to the outputs, we chose to overestimate the prediction uncertainties
- Uncertainties in the SSP climate projections are not available, and so the future prediction uncertainties are underestimated

After transforming the predicted log flow values back into linear flow values, the difference between the inflow and outflow was used to estimate the change in reservoir water level, as discussed previously. The uncertainties were added in quadrature.

To calculate the cumulative predicted reservoir change, the cumulative sum of these monthly changes was taken. The uncertainties were again added in quadrature.

Adding uncertainties in quadrature assumes the uncertainties are independent, however, it allows for a reasonable estimate of the propagation of uncertainty along the future predictions. It should be noted that we chose to limit the future predictions to 10 years because of the reasons discussed above - we simply do not trust that the uncertainty estimations are reliable the deeper into the future we predict.



Predictions for future SRP reservoir water storage changes from present, using two SSP climate projections. Significant decreases in water availability are predicted over the next 10 years.

It should be noted that the difference in start years between the two rivers is due to missing entries in the Salt River historical flow data. The large spike in the SSP2-4.5 predictions for the Verde river appears to be due to a large rainfall event; the spike appears even when training an alternative model to make predictions, so it is unlikely to be an artifact of the model, rather one of the data.

The total capacity of the SRP reservoirs is approximately 3.2 million acre-feet. Our predictions show a decrease in water storage representing a significant fraction of that total capacity by 2036. The rate of decrease far outstrips the rate seen between 2000 and 2020, indicating a worsening of the overall water situation in the immediate future.

Limitations and next steps

Limitations

- We did not have data on reservoir water levels. Therefore, our models can only predict the change in water levels, not the absolute amount.
- We assumed that the relationship between features and targets remain constant over time. In reality, these relationships can change. For example, in recent years, the residents of Phoenix have become more aware of water conservation, resulting in lower water use per person (<https://www.phoenix.gov/administration/departments/waterservices/supply-conservation/save-water/historical-population.html>). Due to this, municipal water demand has not increased with growing population size.
- Estimates of uncertainty in future projections of features were unavailable.
- Accumulation of compounding uncertainties, as predicted future flow data is used to predict further future flow data, is not well understood and is likely underestimated despite generous choices for individual prediction uncertainties
- Only two climate change scenarios were used.

Suggested improvements

- Directly modelling reservoir water levels.
- Using more direct estimates of water demand, rather than proxies such as population size etc., and/or incorporating data on per-capita water use.
- Incorporating uncertainty in future projections of features.
- Incorporating other climate change scenarios.

Conclusions and policy recommendations

Overall our model suggests that water levels in the Salt and Verde river reservoirs will decline significantly in the near future, which will negatively affect Maricopa county's residents. Climate change appears to be the biggest driver of this decline, while data centers had very little impact. To mitigate negative impacts, we suggest policymakers take steps to conserve water, such as by regulating water use and raising public awareness. Additionally, identifying alternate water sources may also help relieve the demand for river water. Further research and monitoring is also recommended.